UNIVERSITY OF JORDAN

Faculty of Business | Department of Finance

ECONOMETRICS PROJECT

Multiple Regression Analysis & Diagnostic Testing

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Course	Research Methods in Finance
Section	Section 2
Country	Türkiye
Sample Period	1995 – 2024 (30 Annual Observations)
Dependent Variable	GDP Growth Rate (% annual)
Independent Variables	Inflation, Gross Capital Formation, Unemployment
Data Source	World Bank World Development Indicators (WDI)

1. Introduction

Türkiye is one of the largest emerging-market economies in the Middle East and Europe, characterised by a dynamic but periodically volatile macroeconomic environment. Between 1995 and 2024, the country experienced several structural transformations including financial crises in 1994 and 2001, a sustained disinflation programme in the early 2000s, rapid credit-driven growth in the 2010s, and resurgent inflationary pressures exceeding 80% in the post-pandemic period.

This study investigates the determinants of annual GDP growth in Türkiye over the period 1995–2024, employing ordinary least squares (OLS) multiple regression analysis. Three macroeconomic indicators are examined as potential explanatory variables: the consumer price inflation rate, gross capital formation (GCF) as a share of GDP, and the total unemployment rate. The analysis relies on annual data sourced from the World Bank World Development Indicators (WDI) database and is estimated using Microsoft Excel's Data Analysis ToolPak.

Each independent variable is expected to influence economic growth through well-established channels. Inflation erodes purchasing power and undermines investor confidence; gross capital formation proxies domestic investment and productive capacity expansion; and unemployment reflects the utilisation of the labour force. Understanding these relationships has both theoretical and policy relevance for an economy as structurally significant as Türkiye.

2. Literature and Theoretical Background

The theoretical foundation of this study draws primarily on the neoclassical growth framework. Solow (1956) demonstrated that capital accumulation, labour input, and technological progress are the fundamental drivers of long-run output growth. Accordingly, gross capital formation — representing the flow of new physical capital — is expected to exert a positive effect on GDP growth by expanding the economy's productive capacity.

Romer (1990) extended this framework through endogenous growth theory, highlighting the role of investment in knowledge and human capital. While this study does not directly measure these channels, GCF serves as a broad proxy for the overall investment environment.

The link between inflation and growth has been explored extensively. Fischer (1993) showed empirically that high inflation is consistently associated with lower economic growth across countries, operating through channels such as reduced real purchasing power, higher borrowing costs, and increased uncertainty deterring investment. This supports the expectation of a negative coefficient on the inflation variable.

The relationship between unemployment and growth is captured by Okun's Law, which posits a negative relationship: higher unemployment implies underutilisation of labour resources and reduced aggregate demand, thereby constraining output growth. Furthermore, the strong negative correlation observed in the data between inflation and unemployment (−0.739) is consistent with the Phillips Curve framework, which describes a short-run trade-off between the two variables (Phillips, 1958). While this introduces a degree of collinearity between regressors, the correlation remains below the conventional threshold of 0.80.

3. Data and Methodology

3.1 Data

The dataset consists of 30 annual observations for Türkiye spanning 1995 to 2024. All data were obtained from the World Bank World Development Indicators (WDI) database. No missing observations were present in the selected series.

Variable	Indicator Code	Unit	Role

GDP Growth Rate	NY.GDP.MKTP.KD.ZG	% annual	Dependent (Y)
Inflation Rate	FP.CPI.TOTL.ZG	% annual	Independent (X_1)
Gross Capital Formation	NE.GDI.TOTL.ZS	% of GDP	Independent (X_2)
Unemployment Rate	SL.UEM.TOTL.ZS	% of labour force	Independent (X_3)

Table 0: Variable Definitions and Data Sources (World Bank WDI)

3.2 Regression Model

The multiple linear regression model estimated is:

$$\mathbf{GDPGrowth_t = \beta_0 + \beta_1 Inflation_t + \beta_2 GCF_t + \beta_3 Unemployment_t + \varepsilon_t}$$

where β_i are the regression coefficients and ε_t is the stochastic error term. Estimation was performed using the OLS method via Microsoft Excel's Data Analysis ToolPak. Two post-estimation diagnostic tests were conducted: White's (1980) test for heteroskedasticity and the Durbin-Watson (DW) test for first-order serial autocorrelation.

4. Descriptive Statistics

Statistic	GDP Growth (%)	Inflation (%)	GCF (% GDP)	Unemployment (%)
Mean	4.96	31.58	28.65	9.98
Median	5.62	13.73	29.49	10.30
Standard Deviation	4.22	29.53	4.35	1.97
Minimum	-5.46	6.25	19.61	6.495
Maximum	11.81	89.11	38.60	14.03

Table 1: Descriptive Statistics — Türkiye, 1995–2024

GDP growth averaged 4.96% over the sample period, but exhibited considerable volatility (standard deviation: 4.22%), ranging from a trough of −5.46% (2001 financial crisis) to a peak of 11.81% (2021 post-pandemic rebound). The wide dispersion underscores the cyclical and crisis-prone nature of the Turkish economy.

Inflation averaged 31.58%, concealing substantial structural change: very high inflation exceeding 80% in the 1990s gave way to single-digit levels during the 2000–2018 period before surging again to over 70% in 2022. This bimodal distribution drives the large standard deviation of 29.53% and a median (13.73%) well below the mean, indicating a right-skewed distribution. Gross capital formation averaged a stable 28.65% of GDP with a comparatively low standard deviation of 4.35%. Unemployment averaged 9.98%, never falling below 6.5%, pointing to persistent structural labour market challenges.

5. Correlation Matrix

Variable	GDP Growth	Inflation	GCF	Unemployment
GDP Growth (%)	1.000	-0.120	0.412	-0.094
Inflation (%)	-0.120	1.000	-0.190	-0.739
GCF (% of GDP)	0.412	-0.190	1.000	0.260
Unemployment (%)	-0.094	-0.739	0.260	1.000

Table 2: Pearson Correlation Matrix — All Variables

GCF shows the strongest bivariate association with GDP growth (r = 0.412), consistent with the investment-growth nexus of neoclassical theory. Both inflation (r = −0.120) and unemployment (r = −0.094) exhibit weak negative correlations with GDP growth, signalling that both variables act as headwinds to economic expansion but have limited bivariate predictive power.

The most notable pairwise correlation is between inflation and unemployment (r = −0.739), reflecting a pronounced Phillips Curve dynamic in the Turkish economy. While this raises a potential multicollinearity concern, the coefficient remains below the conventional critical threshold of 0.80. All correlation signs align with established macroeconomic theory, providing a sound foundation for the regression model.

6. Multiple Regression Results

6.1 Model Fit

Statistic	Value
Multiple R	0.544
R-Squared	0.296
Adjusted R-Squared	0.215
Standard Error	3.739
Observations	30

Table 3: Regression Summary Statistics

The model's R-squared of 0.296 indicates that approximately 29.6% of the variation in Türkiye's annual GDP growth is explained jointly by inflation, GCF, and unemployment. While modest, this level is broadly acceptable for macroeconomic time-series regressions where numerous unobserved shocks inevitably contribute to output fluctuations. The adjusted R-squared of 0.215 confirms that the explanatory contribution of the regressors is genuine rather than an artefact of overfitting.

6.2 GDP Growth Rate vs. Predicted — Chart 1

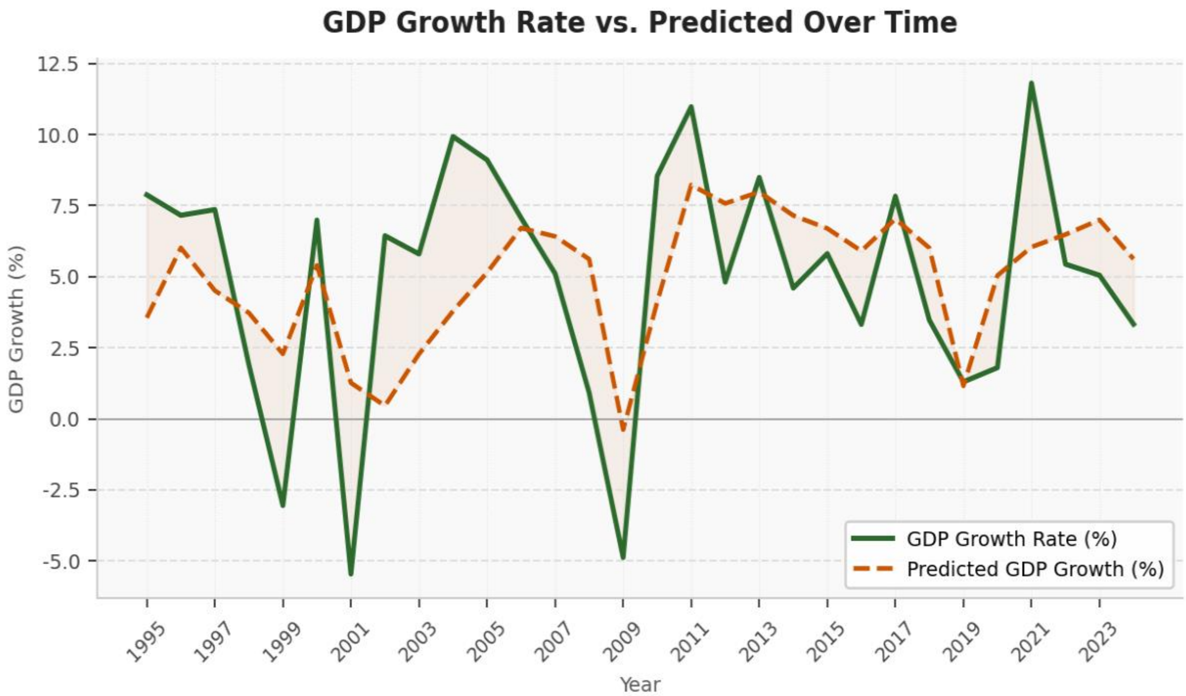

Figure 1: Actual vs. Predicted GDP Growth Rate, Türkiye 1995–2024

Figure 1 plots actual GDP growth (solid green line) against the model's fitted values (dashed orange line). The predicted series tracks the broad cyclical direction of actual growth — capturing the downturns of 1999–2001 and 2008–09 and the recovery phases — but substantially understates peak values such as 2004–05 and 2021. This visual reinforces the moderate R-squared: the model identifies directional trends but cannot fully account for the magnitude of extreme growth episodes driven by idiosyncratic shocks.

6.3 ANOVA and Overall Significance

Source	df	SS	MS	F	Sig. F
Regression	3	152.879	50.960	3.645	0.026
Residual	26	363.530	13.982		
Total	29	516.408			

Table 4: Analysis of Variance (ANOVA)

The F-statistic of 3.645 ($p = 0.026$) is statistically significant at the 5% level. The null hypothesis that all slope coefficients are jointly equal to zero is therefore rejected, confirming that the model as a whole has statistically significant explanatory power.

6.4 Variables Over Time — Chart 2

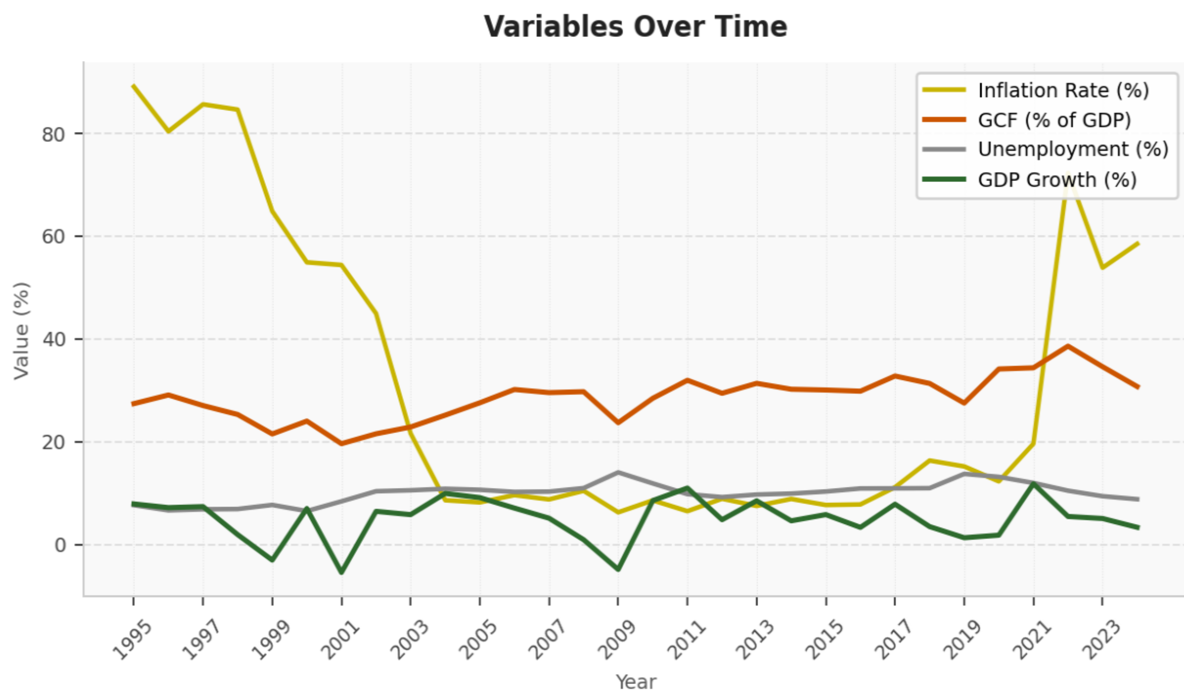

Figure 2: Independent Variables and GDP Growth Over Time, Türkiye 1995–2024

Figure 2 illustrates the temporal evolution of all four variables. The dramatic decline in inflation from above 80% in the late 1990s to single digits by the mid-2000s — and its subsequent resurgence post-2021 — visually explains why inflation's estimated coefficient is statistically weak: its relationship with growth changes character across sub-periods. GCF's gradual upward drift mirrors the economy's investment-led expansion through the 2010s. Unemployment rose sharply following the 2009 global financial crisis and has remained structurally elevated.

6.5 Coefficient Estimates

Variable	Coeff.	Std Err	t-Stat	P-value	Low 95%	Up 95%
Intercept	5.172	7.182	0.720	0.478	-9.591	19.936
Inflation Rate (X_1)	-0.061	0.035	-1.744	0.093	-0.133	0.011
GCF % of GDP (X_2)	0.458	0.165	2.766	0.010	0.118	0.798
Unemployment (X_3)	-1.142	0.533	-2.141	0.042	-2.238	-0.046

Table 5: Regression Coefficients and Statistical Significance

Intercept (5.172, $p = 0.478$): Statistically insignificant; reflects baseline estimated growth when all regressors equal zero — a counterfactual without practical meaning in this context.

Inflation Rate (X_1): −0.061, $p = 0.093$ — Each additional percentage point of inflation is associated with a 0.061 percentage point decline in GDP growth, ceteris paribus. The negative sign is consistent with Fischer's (1993) findings. Significance at only the 10% level reflects the structural break in Türkiye's inflation regime across the sample, which attenuates the average coefficient.

Gross Capital Formation (X_2): +0.458, $p = 0.010$ — GCF is the strongest and most statistically significant predictor. Each one percentage point increase in GCF as a share of GDP is associated with a 0.458 percentage point increase in GDP growth at the 1% level. This strongly supports Solow-Romer capital accumulation theory and represents the principal actionable policy lever identified by the model.

Unemployment Rate (X_3): −1.142, p = 0.042 — Unemployment exerts the largest marginal effect in absolute terms. A one percentage point rise in the unemployment rate is associated with a 1.142 percentage point fall in GDP growth (5% significance). This is consistent with Okun's Law and confirms that labour market slack meaningfully constrains economic output in Türkiye.

7. White's Test for Heteroskedasticity

Heteroskedasticity refers to a condition in which the variance of the regression error term is not constant across observations. In time-series models spanning periods of dramatically different macroeconomic volatility — such as high-inflation 1990s versus the stable 2000s in Türkiye — heteroskedasticity is a common concern. White (1980) proposed a general test that regresses the squared OLS residuals on the original regressors, their squares, and all cross-products.

Statistic	Value
Number of Observations (n)	30
Auxiliary Regression R^2	0.625
Test Statistic ($n \times R^2$)	18.764
Degrees of Freedom	9
χ^2 Critical Value (α = 5%)	16.919
Conclusion	REJECT H_0 — Heteroskedasticity detected

Table 6: White's Test Summary — Auxiliary Regression Results

H_0: Homoskedasticity (constant variance of error terms)

H_1: Heteroskedasticity (non-constant variance of error terms)

The test statistic ($n \times R^2$) of 18.764 exceeds the chi-squared critical value of 16.919 at 9 degrees of freedom (α = 5%), leading to rejection of the null hypothesis. Statistically significant evidence of heteroskedasticity is present in the model. Although OLS coefficient estimates remain unbiased, the standard errors are biased and inconsistent, rendering conventional t-tests and F-tests unreliable. The p-values reported in Table 5 should therefore be interpreted with caution. Remedial measures such as heteroskedasticity-consistent (White-robust) standard errors or weighted least squares estimation would be appropriate in a more advanced treatment.

8. Durbin-Watson Test for Autocorrelation

The Durbin-Watson statistic tests for first-order serial autocorrelation in the regression residuals — a critical concern in time-series econometrics. Autocorrelation implies that error terms are not independent across periods, violating the classical OLS assumption and leading to inefficient estimates with biased standard errors.

Statistic	Value
$\Sigma(e_t - e_{t-1})^2$	679.0
Σe_t^2	363.53
D-W Statistic	1.868
dL (Lower Bound, n=30, k=3)	1.214
dU (Upper Bound)	1.650
No Autocorrelation Region	1.650 – 2.350
Conclusion	No autocorrelation (dU < d < 4 − dU)

Table 7: Durbin-Watson Test Results (n = 30, k = 3)

Decision Rule: Since dU (1.650) < d (1.868) < 4 − dU (2.350), the null hypothesis of no autocorrelation is not rejected.

The DW statistic of 1.868 falls well within the no-autocorrelation zone, indicating that the residuals do not exhibit significant serial dependence over time. The OLS estimates retain their efficiency properties with respect to autocorrelation. This result must nonetheless be qualified by the heteroskedasticity finding — standard errors remain problematic due to non-constant variance even in the absence of serial correlation.

Residuals Over Time — Chart 3

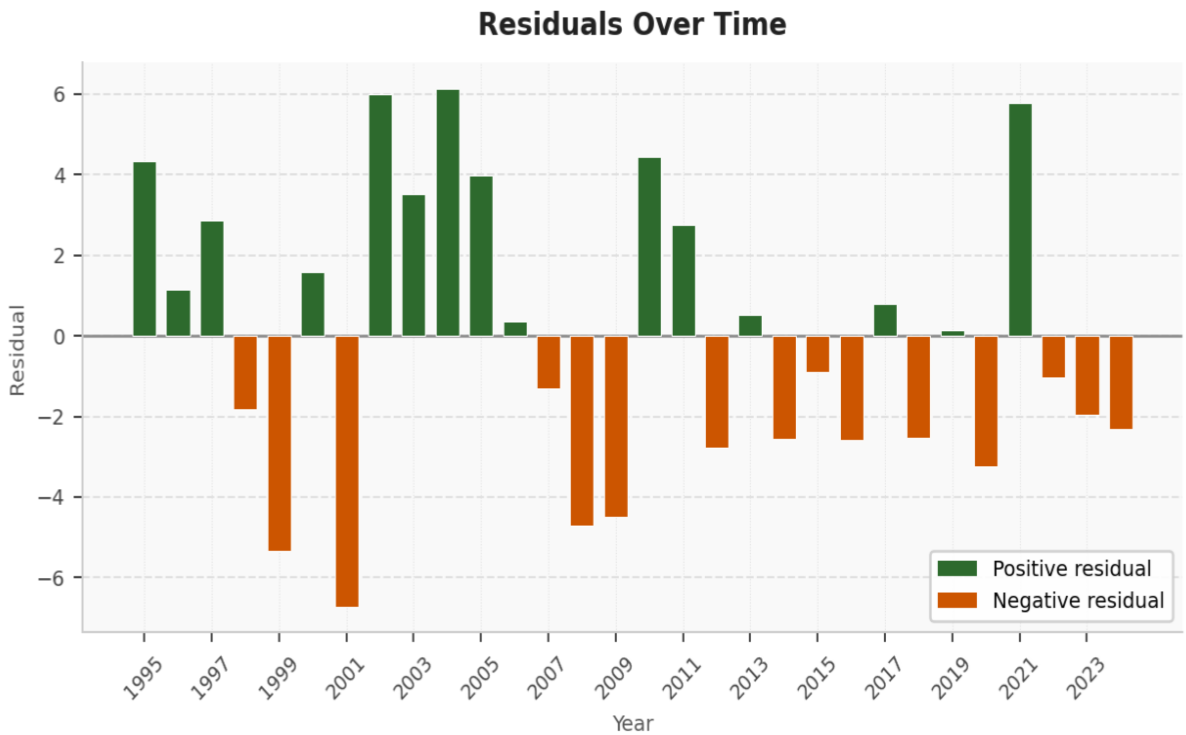

Figure 3: OLS Residuals Over Time, Türkiye 1995–2024

Figure 3 provides a visual diagnostic of the regression residuals. The residuals appear to fluctuate around zero without a systematic trend or persistent pattern, consistent with the Durbin-Watson result of no significant autocorrelation. However, the larger absolute residuals in the earlier years (1995–2002) and in 2021 suggest that error variance may be higher during extreme macroeconomic episodes, visually corroborating the heteroskedasticity detected by White's test.

9. Discussion and Interpretation

Impact of Independent Variables on GDP — Chart 4

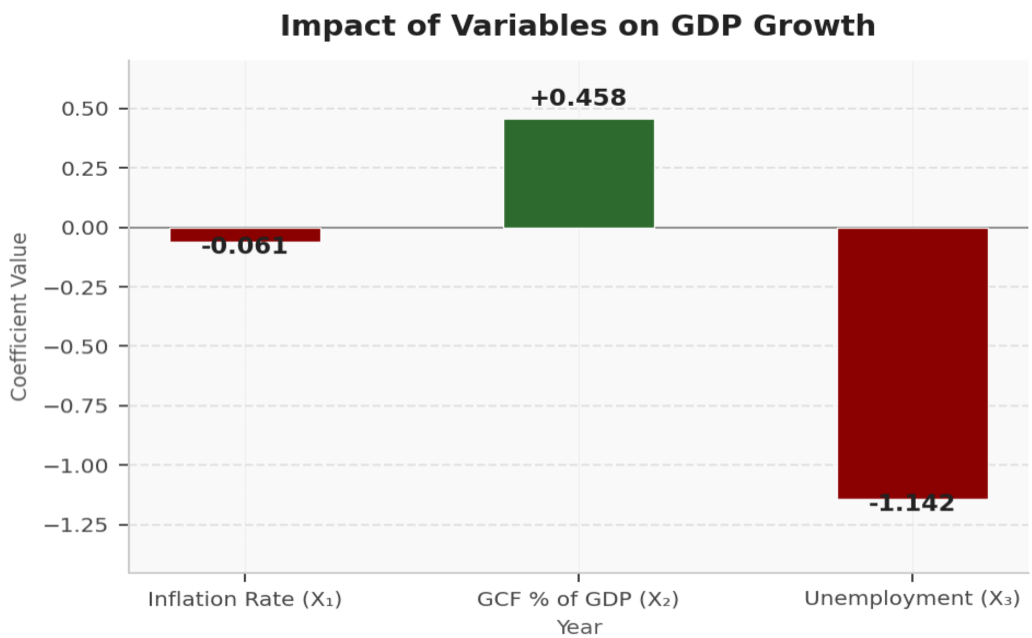

Figure 4: Regression Coefficient Magnitudes — Impact on GDP Growth

Figure 4 visualises the estimated regression coefficients, providing an intuitive summary of each variable's direction and relative magnitude. GCF's positive bar (+0.458) is immediately apparent as the primary growth driver. Unemployment's large negative bar (−1.142) confirms its dominant dampening effect. Inflation's small negative bar (−0.061) captures a real but marginal drag on growth within this model.

Gross capital formation emerges as the most statistically robust and economically significant predictor, confirming the centrality of domestic investment in driving output growth. Each one percentage point increase in GCF as a share of GDP is associated with 0.458 percentage points of additional annual growth. This result carries clear policy implications: sustaining or increasing the investment rate should be a priority for Turkish policymakers.

Unemployment is statistically significant at 5% and carries the largest marginal effect (−1.142). The persistence of unemployment above 6.5% throughout the entire sample is a structural concern that constrains the economy's growth potential. Addressing structural barriers to labour force participation would yield meaningful growth dividends according to the model.

Inflation is statistically significant only at the 10% level, suggesting that its direct growth-dampening effect is present but comparatively weak across the full sample. This may partly reflect the economy's adaptation to chronic inflation during the 1990s and the compressing effect of averaging across two very different inflation regimes. Nonetheless, the negative sign aligns with Fischer (1993) and justifies continued commitment to price stability.

10. Model Limitations and Improvement Suggestions

10.1 Identified Limitations

- Omitted Variable Bias: The model excludes potentially important determinants such as real exchange rates, government expenditure, trade openness, interest rates, and institutional quality. Omitting these variables biases coefficient estimates if they correlate with included regressors.
- Heteroskedasticity: White's test confirms non-constant error variance, likely driven by Türkiye's structural shift from a high-inflation to a low-inflation regime. This renders standard OLS inference unreliable.
- Structural Breaks: Major events — the 2001 Turkish financial crisis, the 2008–09 global financial crisis, and the COVID-19 pandemic — create regime changes that a simple linear OLS model cannot capture without dummy variables or time-varying parameter methods.
- Limited Sample: With $n = 30$ annual observations, statistical power is limited, and adding further variables quickly exhausts degrees of freedom.
- Multicollinearity: The moderate correlation (−0.739) between inflation and unemployment may inflate standard errors for these variables, partially explaining the marginal significance of the inflation coefficient.

10.2 Potential Improvements

- Apply heteroskedasticity-consistent (White robust) standard errors to obtain valid inference under non-constant variance.
- Introduce structural break dummies for the 2001 crisis, 2008–09 global recession, and 2020 pandemic to control for regime shifts.
- Expand the variable set to include the real effective exchange rate, government fiscal balance, and trade-to-GDP ratio to reduce omitted variable bias.
- Consider error-correction or VAR frameworks to better capture dynamic relationships among macroeconomic variables in time series.
- Use logarithmic transformations where appropriate to stabilise variance and improve linearity across the sample.

11. Conclusion

This study employed OLS multiple regression analysis to examine the macroeconomic determinants of GDP growth in Türkiye over 1995–2024, utilising annual World Bank WDI data. Three independent variables were examined: inflation, gross capital formation, and unemployment.

The overall model is jointly statistically significant ($F = 3.645$, $p = 0.026$), explaining approximately 29.6% of GDP growth variation. Gross capital formation is the strongest predictor ($p = 0.010$, coefficient +0.458), confirming that investment drives growth in line with neoclassical theory. Unemployment is significant at 5% (coefficient −1.142), consistent with Okun's Law. Inflation exerts a negative but marginally significant effect ($p = 0.093$), consistent with Fischer's (1993) evidence.

White's test confirms heteroskedasticity (test statistic 18.764 > χ^2 critical 16.919), which calls for robust standard error estimation. The Durbin-Watson statistic of 1.868 indicates no significant serial autocorrelation, supporting the temporal independence of residuals. All coefficient signs conform to economic theory.

In conclusion, investment expansion and labour market improvement are the most impactful policy levers for promoting sustained GDP growth in Türkiye. The model provides a sound econometric baseline; future extensions incorporating additional variables, robust estimation, and structural break adjustments would substantially improve its explanatory power and policy relevance.

12. References

Fischer, S. (1993). The role of macroeconomic factors in growth. Journal of Monetary Economics, 32(3), 485–512. https://doi.org/10.1016/0304-3932(93)90027-D

Phillips, A. W. (1958). The relation between unemployment and the rate of change of money wage rates in the United Kingdom, 1861–1957. Economica, 25(100), 283–299. https://doi.org/10.1111/j.1468-0335.1958.tb00003.x

Romer, P. M. (1990). Endogenous technological change. Journal of Political Economy, 98(5), S71–S102. https://doi.org/10.1086/261725

Solow, R. M. (1956). A contribution to the theory of economic growth. The Quarterly Journal of Economics, 70(1), 65–94. https://doi.org/10.2307/1884513

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. Econometrica, 48(4), 817–838. https://doi.org/10.2307/1912934

World Bank. (2025). World Development Indicators. The World Bank Group. https://databank.worldbank.org/source/world-development-indicators